S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

MERIDIAN

Read the news.
Price the risk.

An explainable NLP risk engine that turns financial news and tweets into risk signals, then rebalances an index and stress-tests a bank book.



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Risk moves with the news. Risk teams read it too late.

The problem

- A sanction, a default or a war changes market and credit risk on the day of the news, before it shows in prices or ratings.
- The news arrives as text: 69,000 articles and posts in one year for only 16 stocks.
- A risk score that nobody can explain does not pass model-risk review. Black-box output is hard to act on.

Our approach

ENGINE
news + tweets → sentiment, event class, impact

→

MODULE A
sentiment-tilted index

+

MODULE B
event-driven stress test

- **No LLM.** Classic NLP models: fast, reproducible, and every score lists the words and parts behind it.
- Both modules from the brief, driven by the same three signal fields.
- A Bloomberg-style terminal that replays the Russia–Ukraine crisis, 17 Feb – 9 Mar 2022.

ONE SIGNAL · GDELT · 21 FEB 2022

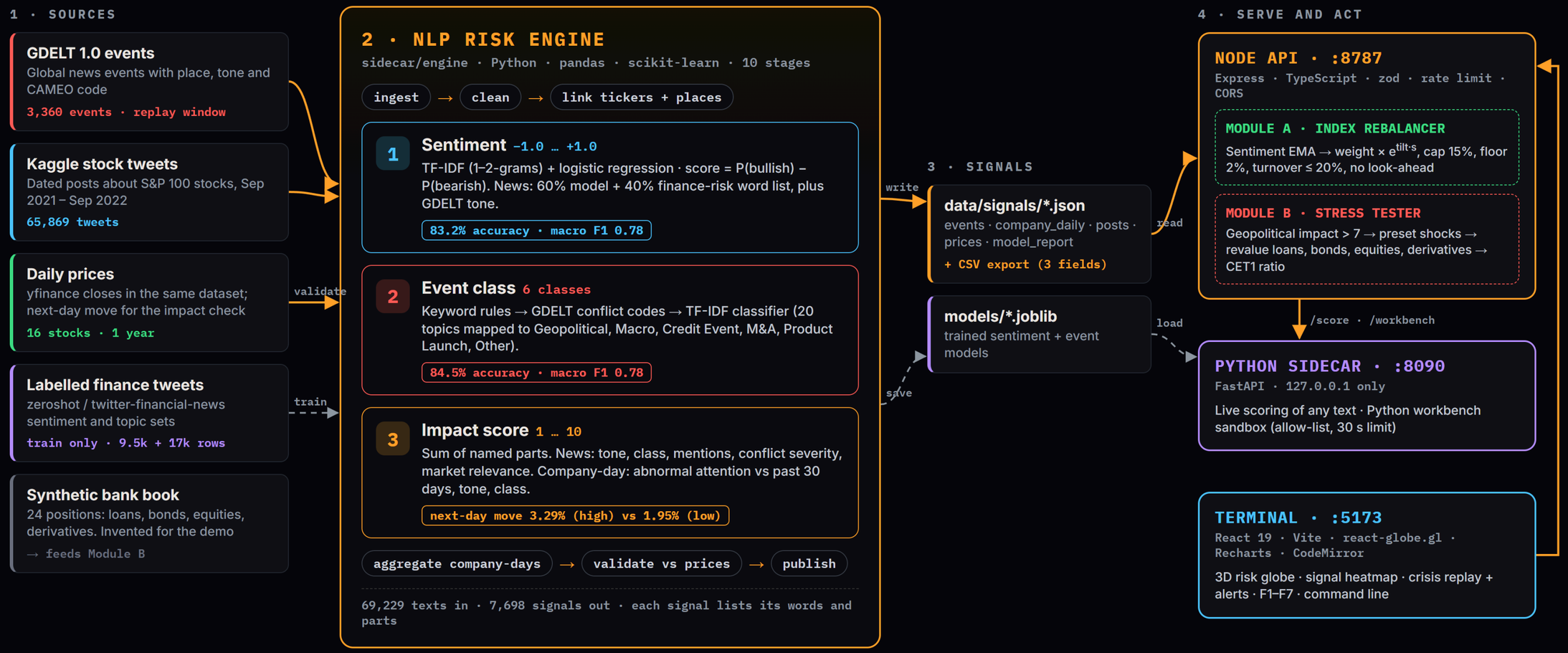
"EU issues massive sanctions warning to Russia"

```
SENTIMENT -0.62
EVENT Geopolitical (rule: "sanctions", "russia")
IMPACT 9 = tone 1.54 + class 3.0 + volume 2.0 + severity 0.72 + relevance 1.5
```

MERIDIAN

SYSTEM ARCHITECTURE · NLP RISK ENGINE → RISK SIGNALS → MODULES A + B

no LLM · every score explained



runtime data ----▶ training / model files

RESULTS sentiment 83.2% · events 84.5% · impact check 1.7× · CET1 13.9% → 10.4% (preset GEO)

Python engine writes signal files · Node API serves them and runs Modules A + B · FastAPI sidecar scores live text

Three explainable models. Two modules on top.

Sentiment `-1 … +1`

TF-IDF 1–2-grams + logistic regression on 9.5k labelled finance tweets. Score = P(bullish) − P(bearish). News adds a finance-risk word list (40%) and GDELT tone.

WHY 68k texts scored in 7 s on a laptop CPU; each word's push on the score is visible.

Event class `6 classes`

Three layers: keyword rules → GDELT CAMEO conflict codes → TF-IDF classifier (20 topics mapped to Geopolitical, Macro, Credit Event, M&A, Product Launch, Other).

WHY rules give high precision on rare, high-stakes classes; the model covers the rest.

Impact `1 … 10`

A sum of named parts. News: tone, class, mentions, conflict severity, market relevance. Company-day: abnormal attention vs the past 30 days, tone, class.

WHY each part is shown; weights were checked against next-day price moves.

Module A · index rebalancer

16 S&P 100 stocks. Sentiment EMA → weight × $e^{\text{tilt}\cdot s}$; cap 15%, floor 2%, turnover ≤ 20%; uses only past signals.

Module B · stress tester

Geopolitical impact > 7 → preset shocks. Loans (PD × LGD), bonds (duration, convexity, spreads), equities (beta), derivatives → CET1.

Stack

Python, pandas, scikit-learn, FastAPI · Node, Express, TypeScript, zod · React 19, Vite, react-globe.gl, Recharts.

Better than naive baselines. Checked on held-out data.

83.2%

sentiment accuracy, 2,388 held-out tweets

vs 65.6% always "neutral"

84.5%

event-class accuracy, 4,117 held-out tweets

vs 45.3% always "Other"

1.7×

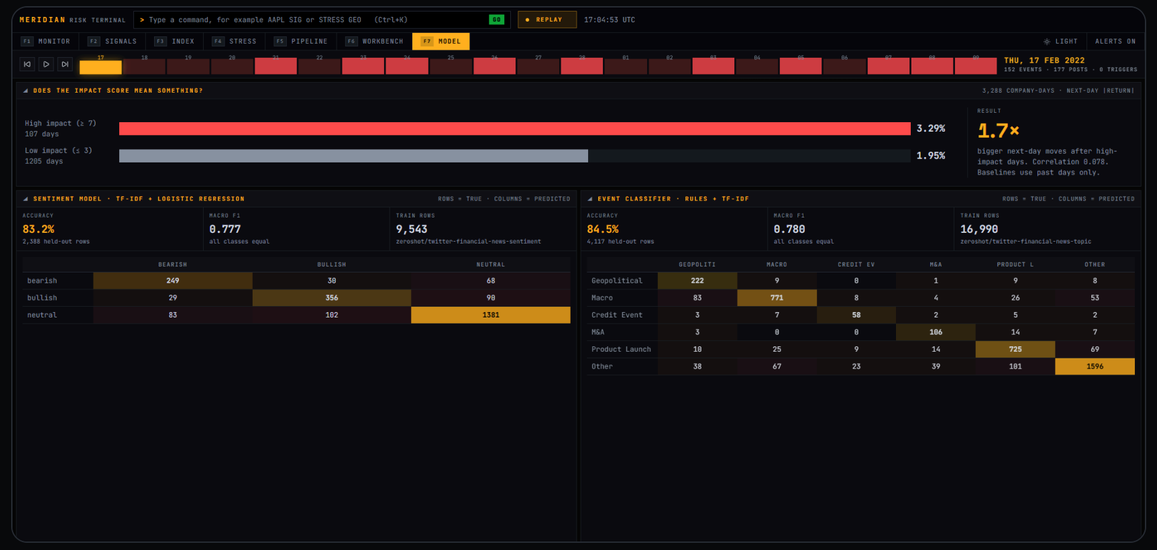
larger next-day move after high-impact days

3.29% vs 1.95% · past data only

10.4%

CET1 after the Geopolitical preset (from 13.9%)

–$196.1M · severe: 9.0%





69,229 texts → 7,698 signals · all 10 pipeline stages logged with row counts and timings

06 · DOMAIN IMPACT

From headline to capital ratio on the day of the event.

Early warning

In the replay, Meridian raised its first Geopolitical alert on 18 Feb 2022 and its first severe one (impact 9) on 21 Feb, three days before the invasion on 24 Feb.

Capital view, not only a score

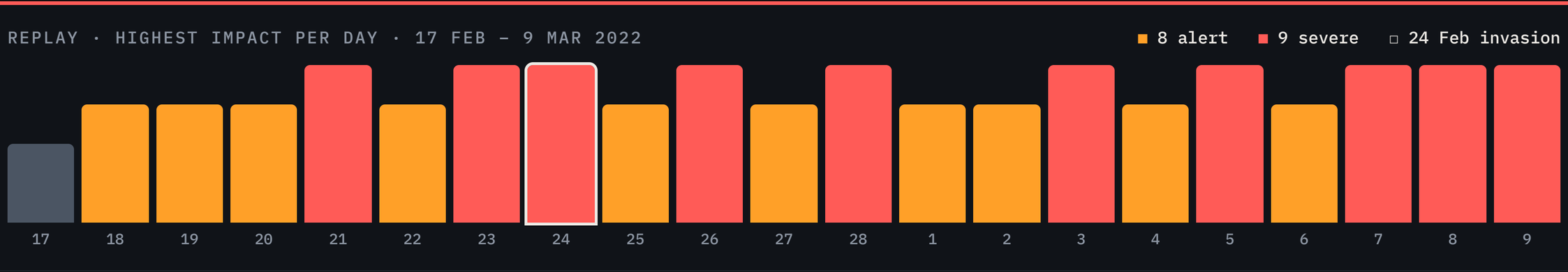
The alert links straight to a stress test: value before and after, P&L by risk factor and the CET1 ratio. A credit or market-risk team sees what the news costs.

Auditable by design

Every sentiment, class and impact value shows its reasons: words, rules, GDELT codes, named parts. This fits model-risk governance, where a black box needs heavy validation.

Cheap and fast to run

No GPU, no LLM API cost, deterministic output. The full pipeline runs on a laptop; live scoring of a new headline takes milliseconds.



Users: market risk · credit risk · index and portfolio teams

What is honest today. What comes next.

Limitations

- Historical sources; the replay covers 17 Feb – 9 Mar 2022.
- GDELT gives no headline text; headlines come from words in the article URL.
- The tilted index did not beat equal weight over the year (−0.98% active). It is a risk view, not a trading signal.
- The bank book and shock sizes are synthetic examples.
- Impact separates high from low days, but its correlation with next-day moves is weak (0.08).

Next steps

- Live feeds: GDELT 15-minute updates and NewsAPI headlines.
- FinBERT as a challenger model, behind the same explanation layer.
- Build the Module B book from the open banking transaction data.
- Calibrate shock sizes to history (2008, 2020, 2022).
- Run each workbench job in its own container without network access.